Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*AI and Vision based Robot Machining and Collision Avoidance for Shipbuilding Industry*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		*Mechanical・Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	*Robot Control and Intelligence Technologies / Robot Vision and Production Automation Technologies*	*Shipbuilding/ Marine Systems*	*Ship Production Systems/Construction Methods*
Concept of Proposed Technology	*Enabling adaptive robot machining and collision avoidance by vision-based robot control, AI based feature detection and path planning and vision-AI-based workspace monitoring for collision partner detection for large parts and especially shipbuilding industry.*			
Rationale for Joint R&D Project	*This project will develop the basis for autonomous machining with robots in shipbuilding, enabling workerless shifts, cost reduction, automation and higher process reliability. Collaboration with Korean partners is required regarding technical requirements, boundary conditions, test and demonstration. Furthermore, to develop concept for further industrial use in Korea.*			
Key Words	1. Artificial intelligence 2. Optical feature detection			

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	3. Robot-based machining 4. Collision avoidance 5. Shipbuilding

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3-4)	End of the Project	Level (7)
R&D Objectives	Enabling adaptive robot machining and collision avoidance in shipbuilding by • Vision-based robot control in mechanical machining • Vision-based feature detection and AI-based path planning, especially for weldseam machining, deburring and cast-iron machining • Vision-based workspace monitoring for AI-based collision partner detection • Approaches for further use of vision systems and AI in machines and robots			
R&D Target Specifications	• Robot machining accuracy <20 µm • Addressed technologies: weldseam, cast iron machining, deburring • Enabling worker-free shifts • Collision reduction >95%			
R&D Plan	1. Analysis of partner needs and customer needs, definition of R&D constraints, detailing project goals in shipbuilding 2. Development of vision-based part and feature detection 3. Development of AI-based deburring strategy 4. Development of AI-based path planning for weldseam and cast-iron machining 5. Development of vision-based workspace monitoring 6. Development of AI-based collision partner detection strategy 7. Development of collision avoidance control strategy 8. System integration and demonstration 9. Development of approaches for further use of vision systems and AI in machines and robots 10. Knowledge transfer Milestone 1: Strategy for AI-based deburring available Milestone 2: Strategy for AI-based cast-iron machining available Milestone 3: strategy for AI-based collision partner detection available Project duration: 3 years			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Consultancy of the Korean partner • Strategy developments with Korean partner • Solution development • System integration and commissioning with Korean partner • Execution of tests and demonstration • Know-how transfer
Expected R&R of Korean SME Partner	• Providing use cases and customer requirements • Supporting strategy development • Providing machining robot for demonstration • Supporting system integration and demonstration • Interest of utilizing results in own products and/or services

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The outcomes and impact of the R&D results will be • Vision-based robot control in mechanical machining in shipbuilding industry • Vision-based feature detection and AI-based path planning for deburring, weldseam and cast-iron machining • Vision-based workspace monitoring for AI-based collision partner detection • Approaches for further use of vision systems and AI in machines and robots • Automation of production • Enabling workerless shifts • Significant reduction of costs in manufacturing For the Korean SME: ➢ Application area: machining of large parts like in shipbuilding industry ➢ To be used in own manufacturing and/or ➢ New products for technology providers

6. References

(1)

(2)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization